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Francesco Sannicola

Machine Learning Engineer — Data Scientist

Working experience

Data Scientist II — ***Agile Lab*** Turin - Remote 09/2025– current

- Engineered a geospatial routing engine on a multi-layer graph for infrastructure planning, with dynamically generated filters, supported by spatial indexing and multi-level **caching**—reducing worst-case query time from 2 minutes to 30 seconds.

Machine Learning Engineer - Senior Consultant — ***Links M&T*** Bari - Hybrid 01/2025– 08/2025

- Built multi-tenant IoT energy forecasting platform for 3 clients. Used genetic algorithms, **MLOps** (tracking, registry, CI/CD), and real-time dashboards. Cut peak costs, consumption, and CO_2 via per-tenant demand forecasts.

Machine Learning Engineer - Consultant — ***Links M&T*** Bari - Hybrid 03/2023–12/2024

- Integrated browsing, crawling, GraphRAG into **ReAct agent chatbot**. Unified on-premise/online sources; cut reporting prep by 15–30 min/session.
- Designed ESG reporting with **RAG** + **GPT**. Automated KPI extraction from fund reports; lowered compliance costs.
- Developed an **NLP** web application powered by **encoder-only transformers** for contextual content recommendations with sub-5s latency, including speech-to-text features.
- Engineered **predictive maintenance** for EV chargers. Degradation models cut costs 30%, prevented failures.
- Developed loan **default prediction** model (80% accuracy, <2% false positives). Flagged risks 3–6 months early.
- **Lecturer** in software development, databases, and Python at *Associazione Formazione*, and in anomaly detection and time series analysis at *Links Academy – AI & ML*. Resulted in 7+ student job placements; practical lab materials available on *GitHub*.

Artificial Intelligence and Data - Analyst — ***Deloitte*** Bari - Remote 10/2022–02/2023

- Developed **ETL** workflow with auto KPI visualizations from surveys. Turned static reports into real-time dashboards for engagement/retention tracking.

AI engineer and System Administrator - Intern — ***Intesa Sanpaolo*** Turin - Hybrid 12/2021–05/2022

- Implemented real-time **time series forecasting** + anomaly detection. Advanced from stats to RNNs/autoencoders; cut failure response from minutes to seconds.

Education

Master's degree in Computer Science - AI — *University of Turin* **GPA**: 3.71/4 01/2020–04/2022

- Coursework: Neural Networks; Machine Learning; Natural Language Technologies; Advanced Software Architectures.
- 200h collabs: web pages development for freshman orientation; study support for a student with disability.
- Thesis: real-time **anomaly detection** using stats, ML, DL (thesis link).

Bachelor's degree in Computer Science — *University of Bari* **GPA**: 4/4 10/2016–12/2019

- Coursework: Computer Architecture; Operating Systems; Algorithms; Databases; Software Engineering.
- Thesis: full-stack indoor positioning system integrating a content-based **recommender system**.

Skills

Programming	Python, Java, C, SQL, Bash
ML/AI	GitHub Copilot, TensorFlow, Keras, PyTorch, scikit-learn, RAG, ReAct agents, LangChain, LangGraph, Transformers
Analytics	Pandas, MS Excel, Qlik Sense, Qlik NPrinting
Databases	PostgreSQL, MongoDB, Redis, Milvus
Cloud & DevOps	AWS (EC2, S3, Lambda, SageMaker, Transcribe, Polly, SQS, SNS, RDS, EventBridge), GCP, Docker, Kubernetes, Git, GitLab CI/CD
Tools	FastAPI, Spring Boot, Pytest, Flask, Keycloak, Uv, Ruff, Pre-commit, Bokeh

Certifications

04/2024	**AWS Certified Machine Learning**	03/2024	AWS Machine Learning Cert Preparation
05/2022	GCP Big Data and ML Fundamentals	04/2022	GCP Fundamentals: Core Infrastructure

Languages

Italian: Native

English: Professional Working Proficiency